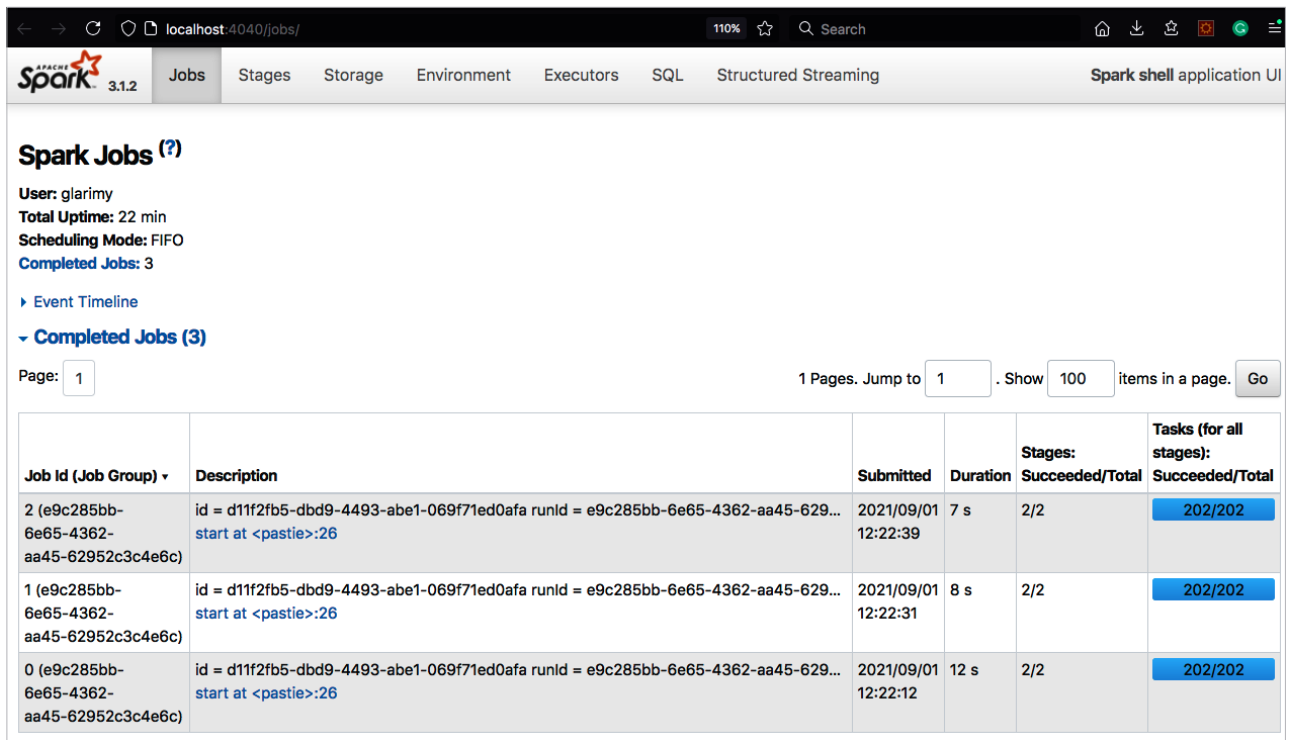




Spark Jobs (?)

User: glarimy
 Total Uptime: 22 min
 Scheduling Mode: FIFO
 Completed Jobs: 3

Event Timeline

Completed Jobs (3)

Page: 1 1 Pages. Jump to 1 . Show 100 items in a page. Go

Job id (Job Group) ▾	Description	Submitted	Duration	Stages: Succeeded/Total	Tasks (for all stages): Succeeded/Total
2 (e9c285bb-6e65-4362-aa45-62952c3c4e6c)	id = d11f2fb5-dbd9-4493-abe1-069f71ed0afa runId = e9c285bb-6e65-4362-aa45-629... start at <pastie>:26	2021/09/01 12:22:39	7 s	2/2	202/202
1 (e9c285bb-6e65-4362-aa45-62952c3c4e6c)	id = d11f2fb5-dbd9-4493-abe1-069f71ed0afa runId = e9c285bb-6e65-4362-aa45-629... start at <pastie>:26	2021/09/01 12:22:31	8 s	2/2	202/202
0 (e9c285bb-6e65-4362-aa45-62952c3c4e6c)	id = d11f2fb5-dbd9-4493-abe1-069f71ed0afa runId = e9c285bb-6e65-4362-aa45-629... start at <pastie>:26	2021/09/01 12:22:12	12 s	2/2	202/202

Figure 7: Admin interface

Machine learning with Spark

Another interesting part of the Spark ecosystem is MLib for machine learning. Data scientists can focus on data processing while Spark takes care of the infrastructure. MLib interoperates with Numpy in Python and R libraries. Any Hadoop data source like HDFS and Hbase can be used to fit into Hadoop workflows, if these exist.

MLib supports all major algorithms like classification, regression, clustering, decision trees, etc. It also offers utilities for constructing, evaluating, saving and loading models and pipelines.

Admin interface

Everything in Spark runs as a job. As noted earlier, the job is split into tasks that carry out in-memory transformations and actions, in parallel, on the workers across the cluster. This model gives Spark an edge in parameters like performance and throughput.

However, this model also poses a challenge to the administrator. Since the job is running across the cluster and is automatically scheduled by the driver, the administrator will have the additional responsibility of tracking it on not just one machine, but on the entire cluster. The admin interface provided by Spark immensely helps the administrator in this regard.

For example, the admin interface for the Spark shell can be accessed at <http://localhost:4040/jobs/>, which is similar to what is shown in Figure 7. Such an interface is available for every Spark application using which the status of the jobs, storage, and executors can be tracked.

Spark Big Data ecosystem

No organisation possibly uses just one

or other Big Data technology. Often, the CTOs deploy several tools and technologies for various processing needs. Apache Spark is one such tool that can be used for batch processing, stream processing, and machine learning at scale.

Spark can work with other big data technologies like Hadoop data sources, Hadoop cluster managers, Apache Kafka, and Apache Flume, to name a few.

On its website, Apache Spark claims 100 times faster performance compared to the nearest competitor. This speed is attributed to the clusterwide in-memory computation model. With first-class support in Java, Python, Scala, and R, Apache Spark's popularity and adoption are only growing! 

By: Krishna Mohan Koyya

The author is the founder and the principal technology consultant at Glarimy Technology Services, Bengaluru. He has conducted more than 500 programmes on design and architecture to corporate clients in the last 12 years. Prior to that, he held various positions in industry and academia for a decade.